

— INDUSTRY-ACADEMIA CONCLAVE —

SustainAI 2026

Intelligence for a Sustainable Tomorrow

— Innovate | Collaborate | Impact —



AI for Sustainable Development Goals (AI for SDGs)



VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering
and Information Systems

Presents

Event Dates: 6–7 October 2026 • Initial Idea Submission Form

SPES

See · Process · Enhance · Speak

An AI + IoT Wearable Assistant that gives sight a voice

School of Computer Science Engineering and Information Systems, VIT Vellore • Event Dates: 6–7 October 2026

1. TEAM DETAILS

Team Name: Team SPES

Team Leader: Kishore D P

Team Type: External Academic Team

2. TEAM MEMBERS

#	Name	Institution / Department
1	Kishore D P (Team Leader)	RMK Engineering College
2	Tarun L	RMK College of Engineering and Technology

3. IDEA DETAILS

3.1 Title of the Proposed Idea

SPES – See, Process, Enhance, Speak: An AI-Powered Wearable Assistant for the Visually Impaired.

3.3 Sustainable Development Goal (SDG) Addressed

Primary SDG: SDG 10 – Reduced Inequalities

Supporting SDGs: SDG 3 – Good Health & Well-Being; SDG 11 – Sustainable Cities & Communities

SDG Number(s): 10 (primary), 3, 11

3.4 Problem Statement

What is the problem?

Visually impaired people cannot independently perceive their surroundings — they struggle to read printed text, recognise objects and people, navigate unfamiliar routes, and respond to sudden threats. Existing aids are either

single-purpose (a cane, a magnifier) or premium smart-glasses costing lakhs of rupees, leaving most users without an affordable, all-in-one solution.

Who is affected by the problem?

Over 2.2 billion people worldwide live with vision impairment (WHO), including an estimated 34 million+ in India. The elderly, low-income households, and people in developing regions are affected most, as advanced assistive devices remain out of financial reach.

Why is the problem important?

Loss of independent mobility and information access directly drives social and economic inequality — limiting education, employment, and personal safety, and increasing dependence on others. Closing this gap is central to SDG 10 (Reduced Inequalities) and to the dignity and inclusion of the visually impaired.

Current challenges / limitations:

- High cost of existing smart assistive wearables (often Rs.50,000–Rs.3,00,000).
- Most devices do one job only — no single tool combines reading, recognition, navigation and safety.
- Little to no personal-safety / emergency support for a vulnerable, often-targeted group.
- Bulky, non-intuitive interfaces that are hard to use hands-free.

3.5 Proposed AI-Driven Solution

SPES is a low-cost, hands-free wearable that acts as a second pair of eyes. An ESP32-CAM captures the scene and streams it over Wi-Fi to an AI backend powered by Google Gemini, which understands the image and returns a clear, natural spoken description through Bluetooth earbuds. A companion ESP32-WROOM-32D (with an I2S microphone) lets the user simply talk to SPES. Its name is its pipeline — it **Sees** through the camera, **Processes** with AI, **Enhances** raw pixels and speech into meaning, and **Speaks** the result back. Core modes:

- **Read Mode** — “read this” captures and reads printed text aloud (OCR + AI).
- **Describe Mode** — identifies objects, people, obstacles and surroundings in real time.
- **Ask Mode** — a conversational voice assistant that answers questions hands-free, with memory across sessions.
- **Navigate Mode** — turn-by-turn Google Maps directions delivered as voice guidance.
- **Guardian Mode (SOS)** — on an emergency command, SPES captures a photo of the threat, pins the user’s live GPS location, and sends an automated alert with location to a registered emergency contact / responder.

3.6 Innovation / Novelty

What is innovative or different about your idea?

- **All-in-one at ultra-low cost:** reading, scene understanding, navigation AND personal safety in a single wearable built from ~Rs.1,500–Rs.2,000 of hardware and free AI APIs.
- **Generative-AI vision:** uses a multimodal LLM (Gemini) for open-ended scene understanding instead of a fixed, pre-trained object list — it can describe *anything*, not just trained classes.
- **Guardian Mode:** a safety-first feature purpose-built for a vulnerable group — turning the device from an accessibility aid into a personal protector.
- **Conversational memory:** SPES remembers context across sessions, behaving like a personal companion, not a query box.
- **Split “dumb-eye / smart-brain” architecture** keeps the wearable light and cheap while heavy AI runs in the cloud.

3.7 Target Users / Beneficiaries

Who will use or benefit from your solution?

- Blind and visually impaired individuals seeking independence in daily life.
- Elderly people with declining vision.
- Low-income users priced out of premium assistive technology.

- Caregivers, families, NGOs, and schools for the blind.
- Broader hands-free users — SPES also works as a general-purpose AI assistant.

3.8 Expected Impact

- Restores independence, mobility and confidence for the visually impaired.
- Reduces inequality by making advanced assistive tech affordable and accessible (SDG 10).
- Improves personal safety through instant, location-aware emergency alerts.
- Lowers dependence on caregivers and opens access to education and employment.

4. AI COMPONENT

4.1 Proposed AI Technology

Select the technologies that may be used:

☒ Machine Learning

☒ Deep Learning

☒ Generative AI

☒ Computer Vision

☒ Natural Language Processing

☒ Large Language Models

☐ Reinforcement Learning

☐ Predictive Analytics

☐ Time-Series Analysis

☒ AI Agents

Other: Multimodal LLM (Google Gemini), Speech-to-Text & Text-to-Speech, on-device wake-word / voice activity detection.

4.2 Role of AI in the Proposed Solution

- **Scene understanding & OCR:** Gemini (multimodal LLM + computer vision) reads text and describes the environment.
- **Voice interface:** speech-to-text interprets commands; the LLM classifies intent (Read / Describe / Ask / Navigate / SOS) and generates answers.
- **Natural speech output:** text-to-speech converts responses into a natural voice.
- **Agentic control:** an AI-agent layer routes each command to the correct module and maintains conversational memory.

4.3 Data Requirements

What type of data will be required?

☒ Image / Video Data

☒ Sensor / IoT Data

☐ Satellite / Remote Sensing Data

☒ Text Data

☒ Audio Data

☐ Time-Series Data

☒ Public / Open Data

☐ Organization / Industry Data

Potential Data Source(s): Live camera frames from the ESP32-CAM; microphone audio from the user; GPS / location data; Google Maps API for routing; Google Gemini API for vision, language and reasoning. No sensitive personal datasets are required to train the system.

5. PROPOSED SOLUTION FRAMEWORK

Technology / Implementation Plan

Pipeline: ESP32-CAM (See) → Wi-Fi → Backend API → Gemini (Process/Enhance) → TTS → Bluetooth earbuds (Speak). A parallel ESP32-WROOM-32D handles the microphone, Bluetooth audio and voice commands.

- **Hardware:** ESP32-CAM (AI-Thinker) as the camera/eye; ESP32-WROOM-32D as the coordinator/brain; INMP441 I2S MEMS microphone; Bluetooth earbuds; Li-ion + TP4056 + boost/buck power module.

- **AI & APIs:** Google Gemini (vision, OCR, reasoning), Speech-to-Text, edge-tts / Text-to-Speech, Google Maps API.
- **Backend:** Python (Flask) server exposing endpoints (/read, /ask, /navigate, /sos); LiveKit voice pipeline for real-time conversation.
- **Software:** Arduino/C++ firmware for ESP32 boards; Python for the AI backend and orchestration.

Cloud / Deployment Platform: Cloud-hosted Python backend (initially laptop/LAN for the prototype; deployable to any low-cost cloud VM). AI inference via Google Gemini APIs.

Database / Backend: Python (Flask) REST backend; lightweight JSON / local store for conversational memory and user settings.

Other Technologies: I2S audio, Bluetooth A2DP, Wi-Fi (2.4 GHz), GPS, Google Maps Directions API.

7. FEASIBILITY

7.1 Technical Feasibility

The core system is already working. A full software prototype runs today: voice command → AI intent classification → Read (OCR + speech), Ask (Q&A + speech) and conversational memory, using a laptop camera/mic as temporary stand-ins. The ESP32-CAM has been flashed and verified, the ESP32-WROOM-32D brain (Wi-Fi + Classic Bluetooth) is confirmed, and a Flask backend with /read and /ask endpoints is tested. Remaining work is hardware integration (final camera module + microphone) and adding the Navigate and Guardian modules — all built on proven, available APIs.

7.2 Resource Requirements

- **Hardware:** ESP32-CAM, ESP32-WROOM-32D, OV2640 camera, INMP441 mic, Bluetooth earbuds, Li-ion cells + TP4056 + boost/buck (~Rs.1,500–Rs.2,000 total).
- **Software / APIs:** Google Gemini API (free tier), Google Maps API, Python, Arduino IDE — all free or low-cost.
- **Infrastructure:** a basic cloud VM (or laptop for the prototype) and a 2.4 GHz Wi-Fi / mobile hotspot connection.

7.3 Scalability

SPES scales cleanly: the wearable stays cheap because all heavy AI runs in the cloud, so improving the AI upgrades every device at once. The same platform can add languages, currency/medicine recognition, indoor navigation, and integration with city emergency services — enabling deployment through NGOs, government accessibility schemes and schools for the blind across regions.

8. EXPECTED OUTCOME / PROTOTYPE

A working wearable prototype that, on voice command, reads text aloud, describes surroundings, gives turn-by-turn voice navigation, and triggers a location-aware emergency SOS — all hands-free through earbuds. A software demo already demonstrates the See–Process–Enhance–Speak pipeline end to end.

9. COMPETITIVE ADVANTAGE

What makes your solution valuable compared with existing solutions?

Feature	Premium smart glasses	Basic aids	SPES
Cost	Rs.50k–Rs.3L	Low	~Rs.2k
Scene understanding (open-ended AI)	Limited	No	Yes
Reads any text (OCR)	Some	No	Yes
Voice navigation	Some	No	Yes
Emergency SOS (Guardian Mode)	No	No	Yes
Conversational assistant + memory	No	No	Yes

10. FUTURE POTENTIAL

- Multi-language and regional-language support for wider reach.
- Currency, medicine-label and face recognition for trusted contacts.
- Indoor navigation and public-transport guidance.
- Direct integration with city emergency services (112) and family apps.
- Commercialization via NGOs, CSR programs and government accessibility schemes; potential for incubation and social-enterprise scaling.

11. TEAM CAPABILITY

The team combines embedded-systems and AI-software skills. It has hands-on experience with ESP32 firmware (camera, I2S audio, Bluetooth, Wi-Fi), Python backend development (Flask, REST APIs), and applied AI using the Google Gemini multimodal API, speech-to-text and text-to-speech pipelines. The working software prototype built so far demonstrates the team's ability to take SPES from concept to a functional system.

12. REFERENCES / EXISTING WORK

- WHO — *World Report on Vision / Blindness and Vision Impairment* fact sheets (prevalence data).
- Google Gemini API — multimodal vision & language documentation (ai.google.dev).
- Google Maps Platform — Directions API documentation.
- Espressif ESP32-CAM & ESP32-WROOM-32D technical datasheets.
- Existing assistive devices reviewed: OrCam MyEye, Envision Glasses, Be My Eyes — benchmarked on cost and features.

13. ONE-MINUTE IDEA SUMMARY

Summarize your complete idea in not more than 100 words.

SPES (See, Process, Enhance, Speak) is a low-cost AI + IoT wearable that acts as a second pair of eyes for the visually impaired. Using an ESP32-CAM and Google Gemini AI, it reads text aloud, describes surroundings, and answers questions hands-free through earbuds. It adds Google Maps voice navigation and a Guardian Mode that, in an emergency, captures a photo of the threat and sends the user's live location to a responder. Built from ~Rs.2,000 of hardware and free AI, SPES restores independence, mobility and safety — advancing SDG 10 by reducing inequality for the visually impaired.

14. IDEA PITCH IN ONE SENTENCE

"Our idea uses AI (multimodal LLM + computer vision) and IoT to let the visually impaired see, navigate and stay safe for blind and low-vision individuals, resulting in restored independence, mobility and personal safety."

15. DECLARATION

We hereby declare that the information provided in this submission is true to the best of our knowledge. The proposed idea is our team's original concept and does not knowingly violate the intellectual property rights of any individual, organization, or institution. We agree to participate in the SustainAI 2026 Ideathon and abide by the rules and evaluation procedures specified by the organizers.

Team Leader Name: Kishore D P

Signature: _____

Date: 13-09-2026

FOR ORGANIZER USE ONLY

Submission ID: _____

Date Received: _____

Eligibility Verified: ☐ Yes ☐ No

Submission Complete: ☐ Yes ☐ No

SDG Relevance: ☐ Yes ☐ No

AI Relevance: ☐ Yes ☐ No

Initial Evaluation Score: _____

(out of 100)

Shortlisted: ☐ Yes ☐ No

Evaluator Name: _____

Evaluator Signature: _____

INITIAL IDEA EVALUATION CRITERIA

Criterion	Weightage
Problem Relevance and Clarity	15
Innovation and Novelty	20
AI Integration	15
Feasibility	15
Sustainability / SDG Impact	15
Scalability and Future Potential	10
Clarity of Presentation	10
Total	100

Recommended submission length: 3–5 pages excluding the cover page.